

Strong Stability of Indirect LDDMM Minimizers: the Sharp Positive/Zero Regularization Dichotomy

Automated Math Discovery Candidate Proof

August 11, 2026

Status. Candidate full answer to Remark 7.6 of the source paper, likely valid pending expert review. Strong subsequential stability always holds for the regularized problem $\gamma > 0$. It fails at $\gamma = 0$, which the printed stability theorem includes even though its proof divides by γ .

Source and Open Question

This packet concerns Remark 7.6 on page 15 of Chen and Öktem [1], arXiv:1706.04048v2. In the Hilbert space

$$\mathcal{H} = L^2([0, 1], V),$$

the source minimizes

$$\mathcal{E}_{g,\gamma}(\nu) = \gamma \|\nu\|_{\mathcal{H}}^2 + \mathcal{D}(\mathcal{A}(\nu), g), \quad \mathcal{A}(\nu) := T(\mathcal{V}(\nu, I)). \quad (1)$$

Here $\mathcal{V}(\nu, I)$ is the deformed template, T is the forward operator, and $\mathcal{D}$ is the data discrepancy. Theorem 7.5 proves that for $g_k \rightarrow g$, a sequence of minimizers of $\mathcal{E}_{g_k,\gamma}$ has a weakly convergent subsequence whose limit minimizes $\mathcal{E}_{g,\gamma}$. Remark 7.6 asks whether a strongly convergent subsequence always exists.

Hence, $\mathcal{E}_{g_k, \gamma}(\mathbf{0})$ is a bounded sequence and therefore $\{\nu_k\}_k \subset \mathcal{L}^2([0, 1], V)$ is bounded. Hence, it has a subsequence that converges weakly to some $\hat{\nu} \in \mathcal{L}^2([0, 1], V)$. The continuity of the forward operator along with [Corollary 7.3](#) implies

$$\mathcal{T}(\mathcal{V}(\nu_k, I)) \rightarrow \mathcal{T}(\mathcal{V}(\hat{\nu}, I)) \quad \text{in } Y \text{ as } \nu_k \rightharpoonup \hat{\nu} \text{ in } \mathcal{L}^2([0, 1], V).$$

Since the data discrepancy is continuous,

$$\mathcal{D}(\mathcal{T}(\mathcal{V}(\nu_k, I)), g_k) \rightarrow \mathcal{D}(\mathcal{T}(\mathcal{V}(\hat{\nu}, I)), g) \quad \text{as } g_k \rightarrow g \text{ in } Y.$$

Finally, $\|\hat{\nu}\|_{\mathcal{L}^2([0, 1], V)} \leq \liminf_k \|\nu_k\|_{\mathcal{L}^2([0, 1], V)}$ and ν_k is a minimizer of $\mathcal{E}_{g_k, \gamma}$, so

$$\mathcal{E}_{g, \gamma}(\hat{\nu}) \leq \liminf_k \mathcal{E}_{g_k, \gamma}(\nu_k) \leq \liminf_k \mathcal{E}_{g_k, \gamma}(\nu) \quad \text{for any } \nu \in \mathcal{L}^2([0, 1], V).$$

Hence, $\mathcal{E}_{g, \gamma}(\hat{\nu}) \leq \mathcal{E}_{g, \gamma}(\nu)$ as $g_k \rightarrow g$, but this holds for any $\nu \in \mathcal{L}^2([0, 1], V)$, so $\hat{\nu}$ is a minimizer of $\mathcal{E}_{g, \gamma}$.

In conclusion, the reconstructed velocity fields depend continuously on data in the weak sense, which in turn concludes the proof. ■

Remark 7.6. A natural open question is to study whether the same result holds in the strong sense, i.e., whether there exists a sub-sequence of $\{\nu_k\}_k$ that converges strongly to a minimizer of $\mathcal{E}_{g, \gamma}$.

Next is to consider convergence, again in the weak sense.

Theorem 7.7 (Convergence). *Assume the conditions in [Theorem 7.5](#) holds and there exists some $\nu^* \in \mathcal{L}^2([0, 1], V)$ such that $\mathcal{T}(\mathcal{V}(\nu^*, I)) = g$. Furthermore, let $\{g_k\}_k \subset Y$ be sequence*

The proof of Theorem 7.5 uses the convergence

$$\mathcal{D}(\mathcal{A}(\nu_k), g_k) \rightarrow \mathcal{D}(\mathcal{A}(\hat{\nu}), g) \quad \text{whenever } \nu_k \rightharpoonup \hat{\nu}, g_k \rightarrow g. \quad (2)$$

This follows from [Corollary 7.3](#) and continuity of T when the discrepancy is jointly continuous. We retain precisely this proof-effective continuity hypothesis below.

Proof Intuition

The source proof already contains almost all of the strong-convergence argument. Weak convergence gives the lower bound $\|\hat{\nu}\| \leq \liminf \|\nu_k\|$. To obtain the missing upper bound, use the weak limit itself as a fixed competitor in the minimizing inequality. The discrepancy on both sides converges to the same number by (2), so it cancels in the limit and leaves $\limsup \|\nu_k\|^2 \leq \|\hat{\nu}\|^2$, provided $\gamma > 0$. Weak convergence plus convergence of norms is strong convergence in a Hilbert space.

This mechanism necessarily disappears at $\gamma = 0$. A zero template and zero forward data make the discrepancy independent of the velocity, so every velocity is a minimizer. An orthonormal sequence then gives weak convergence without any strongly convergent subsequence.

Sharp Resolution

Theorem 1 (Strong stability for positive regularization). *Assume the hypotheses and the discrepancy convergence used in the proof of Theorem 7.5 of the source paper. Let $\gamma > 0$, let $g_k \rightarrow g$, and choose*

$$\nu_k \in \operatorname{argmin}_{\nu \in \mathcal{H}} \mathcal{E}_{g_k, \gamma}(\nu).$$

Then some subsequence converges strongly in $\mathcal{H}$ to a minimizer of $\mathcal{E}_{g,\gamma}$.

More precisely, every subsequence satisfying $\nu_k \rightharpoonup \hat{\nu}$ and (2) satisfies $\nu_k \rightarrow \hat{\nu}$ strongly.

Proof. Because $\gamma > 0$, the coercive estimate in the source proof is valid and shows that (ν_k) is bounded. Passing to a subsequence, Theorem 7.5 gives

$$\nu_k \rightharpoonup \hat{\nu} \quad \text{in } \mathcal{H},$$

where $\hat{\nu}$ minimizes $\mathcal{E}_{g,\gamma}$. Put

$$r_k := \mathcal{D}(\mathcal{A}(\nu_k), g_k), \quad r := \mathcal{D}(\mathcal{A}(\hat{\nu}), g).$$

By (2), $r_k \rightarrow r$. Continuity in the data variable also gives

$$\mathcal{D}(\mathcal{A}(\hat{\nu}), g_k) \rightarrow r.$$

Use $\hat{\nu}$ as a competitor for the g_k problem. Minimality gives

$$\gamma \|\nu_k\|_{\mathcal{H}}^2 + r_k \leq \gamma \|\hat{\nu}\|_{\mathcal{H}}^2 + \mathcal{D}(\mathcal{A}(\hat{\nu}), g_k).$$

Taking the upper limit and cancelling the common discrepancy limit yields

$$\limsup_{k \rightarrow \infty} \|\nu_k\|_{\mathcal{H}}^2 \leq \|\hat{\nu}\|_{\mathcal{H}}^2.$$

Weak lower semicontinuity gives the reverse inequality

$$\|\hat{\nu}\|_{\mathcal{H}}^2 \leq \liminf_{k \rightarrow \infty} \|\nu_k\|_{\mathcal{H}}^2.$$

Thus $\|\nu_k\|_{\mathcal{H}} \rightarrow \|\hat{\nu}\|_{\mathcal{H}}$. The Radon–Riesz property of Hilbert spaces now implies $\nu_k \rightarrow \hat{\nu}$ strongly. $\square$

Proposition 1 (Failure at zero regularization). *The strong conclusion in Remark 7.6 is false for $\gamma = 0$ under the printed hypotheses. In fact, the weak compactness conclusion of Theorem 7.5 can also fail at $\gamma = 0$.*

Proof. Choose any nonzero admissible Hilbert space V of vector fields and set $\mathcal{H} = L^2([0, 1], V)$. Take the template $I \equiv 0$. Both the geometric and mass-preserving LDDMM actions keep this template equal to zero under every velocity. Let the data space be $Y = \mathbb{R}$, let $T \equiv 0$, take $g_k = g = 0$, and use the standard squared discrepancy

$$\mathcal{D}(y, g) = |y - g|^2.$$

All continuity and regularity hypotheses in Theorem 7.5 are satisfied. At $\gamma = 0$, however, (1) is identically zero, so every $\nu \in \mathcal{H}$ is a minimizer.

Fix a unit vector field $z \in V$ and let

$$\nu_k(t) = \sqrt{2} \sin(2\pi kt) z.$$

The sequence (ν_k) is orthonormal in $\mathcal{H}$. Hence $\nu_k \rightharpoonup 0$, and 0 is a minimizer, but no subsequence converges strongly because $\|\nu_j - \nu_k\|_{\mathcal{H}} = \sqrt{2}$ for $j \neq k$. This disproves the strong statement at $\gamma = 0$.

Moreover, since every velocity is a minimizer, the sequence $k\nu_1$ is an unbounded sequence of minimizers and has no weakly convergent subsequence. Thus Theorem 7.5 itself requires $\gamma > 0$ (or a separate bounded-selection hypothesis); its displayed coercive estimate contains $1/\gamma$ and is not valid at zero. $\square$

Remark 1 (Exact answer). *For the regularization regime actually supported by the source proof, $\gamma > 0$, Remark 7.6 has an affirmative answer without uniqueness or convexity. For the additional value $\gamma = 0$ allowed by the printed theorem, the answer is negative. This dichotomy is sharp.*

Verification, Scope, and Novelty

The positive proof was checked line by line against page 15 of the source. It uses only the weak subsequence and discrepancy convergence established there, minimality with one fixed competitor, and the Hilbert-space weak-plus-norm criterion. The counterexample uses the source’s own LDDMM state space and the standard squared discrepancy; it is not an abstract pathology.

There are two statement-level points for human review. First, the source prints $\gamma \geq 0$ but divides by γ in its first stability estimate. Second, its theorem states continuity with the limiting datum fixed, while its proof uses continuity along $(\mathcal{A}(\nu_k), g_k)$. Theorem 1 assumes exactly the latter convergence, as the source proof does; joint continuity of $\mathcal{D}$ is a standard sufficient condition.

A bounded novelty check on August 11, 2026 searched the run indexes, exact title and exact-question phrases, and the 32 citing works indexed by OpenAlex. No later paper was found that states this positive/zero dichotomy or closes Remark 7.6. A targeted inspection of the related arXiv:1810.08596 found a weak stability proposition, not the norm-convergence upgrade. Novelty confidence is moderate pending specialist review.

References

- [1] C. Chen and O. Öktem, *Indirect Image Registration with Large Diffeomorphic Deformations*, SIAM J. Imaging Sci. 11 (2018), no. 1, 575–617; arXiv:1706.04048v2, DOI: 10.1137/17M1134627.